ModelIt!

Student Activity Pack

The Ocean Is Turning to Acid

How CO2 Emissions Are Dissolving Marine Life

Grade 09L1 | NGSS HS-ESS3-6, HS-PS1-7 | Systems & Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is the pH scale and what does it measure?

2 Where does the CO_2 from burning fossil fuels go after it enters the atmosphere?

3 What are coral reefs made of and why are they important?

4 How could a gas in the atmosphere affect creatures living deep in the ocean?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

CO2 Emissions | Ocean CO2 Absorption | Ocean pH | Coral Health

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Current Trajectory

CO2 Emissions: HIGH (current rate)

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Paris Agreement

CO2 Emissions: MEDIUM (reduced)

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Zero Emissions

CO2 Emissions: LOW (near zero)

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Water Temperature — Warmer water holds less dissolved CO_2 but accelerates chemical reactions. Rising ocean temperatures interact with acidification in complex ways.

Biological Buffering — Healthy marine ecosystems can partially buffer pH changes through photosynthesis (which consumes CO_2). Losing marine plants reduces this natural buffer.

Deep Ocean Circulation — Ocean currents carry acidified surface water to the deep ocean. This means acidification impacts spread globally, even to remote areas far from emission sources.

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How is the CO_2 from your car slowly dissolving the shells of sea creatures thousands of miles away? Use evidence from your simulation to support your answers.

1. **Why is a 0.1 pH change considered significant when the scale goes from 0 to 14?**

2. **How does the law of conservation of mass explain where atmospheric CO_2 ends up in the ocean?**

3. **Why can't marine organisms simply adapt to more acidic water?**

4. **What makes ocean acidification an especially difficult problem to reverse?**

5. **How does this model connect local actions (driving a car) to global impacts (coral death)?**

8 STEM Challenge: The Coral Reef Rescue Plan

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YOUR ENGINEERING MISSION

Design an intervention strategy that protects a coral reef from acidification while addressing the root cause of CO_2 emissions.

THE SCENARIO

A Caribbean coral reef that supports local fishing and tourism is showing signs of bleaching and dissolution. Your team must propose a multi-pronged strategy: immediate reef protection, medium-term emission reduction, and long-term ocean chemistry restoration.

Engineering Design Thinking

1. WHAT ARE THE MOST URGENT THREATS TO THE REEF RIGHT NOW?
2. HOW WILL YOUR PLAN ADDRESS BOTH SYMPTOMS (ACIDIFICATION) AND CAUSES (EMISSIONS)?
3. WHAT TRADE-OFFS EXIST BETWEEN IMMEDIATE PROTECTION AND LONG-TERM SOLUTIONS?
4. HOW WOULD YOU MEASURE WHETHER YOUR INTERVENTION IS ACTUALLY WORKING?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Marine Chemists and Oceanographers study the chemical composition of seawater and how it's changing. They deploy sensors across the ocean, model chemical reactions at global scale, and advise

governments on climate policy. This field combines chemistry, biology, and environmental science.